Rafael **Avila-Terkenlis**

formerly Rafael **Cottom**

r(dot)cottom(at)rutgers(dot)edu, Rafa-Terkenlis

Ph.D. Student

Rm. 340, Wright-Rieman Laboratories, Busch Campus, Rutgers University–New Brunswick

Interests

Research | Thermophysical evolution of ice-rich planetary surfaces and interiors; cryovolcanism; habitability.

Service | Earth and space science outreach; mentoring students in research and career development.

Education

2030 | **Rutgers University,** New Brunswick, NJ
Ph.D., Earth and Planetary Sciences
Advisor: Lujendra Ojha

2025 | **University of California, Santa Cruz,** Santa Cruz, CA
B.S., Earth Sciences (Planetary Science)

2023 | **Santa Barbara City College,** Santa Barbara, CA
A.S., Astronomy [transfer w/o degree]

Additional Training

2026 | International Heat Flow Commission (IHFC) Summer School | *İzmir, Türkiye*

Student Research Positions

2024-2025 **Undergraduate Researcher** · Advisors: Benjamin Idini & Jonathan Fortney
Solar System Physics (S^2P) Group, Department of Astronomy & Astrophysics, UCSC

2023 **Cosmology Research Intern** · Advisor: Adrian Liu
McGill Cosmic Dawn and Hydrogen Cosmology Group, Trottier Space Institute, Université McGill

2022 **Astrophysics Research Intern** · Advisor: Erika Holmbeck
Carnegie Observatories (Carnegie Astrophysics Summer Student Internship)

Publications

Abstracts

Cottom R. & Ojha L. (2026). Dike-Mediated Water Transport in the Europan Ice Shell: Implications for the Feasibility of Ice-Ocean Exchange. *57th LPSC 3160* (#1881).

Cottom R. & Holmbeck E. (2023). Deriving the First Stellar Mass Function from the Light-Element Abundances of Metal-poor Stars. *Bulletin of the American Astronomical Society, 55*(2).

In Preparation

Avila-Terkenlis R. & Ojha L. (2027). Liquid Water Ascent in the Europan Ice Shell: Plausible Dike-Like Conduit Geometries.

Balkan-Pazvantoğlu E., […], **Avila-Terkenlis R.**, […]. [IHFC Summer School 2026 heat flow data publication].

Honors & Grants

2026 Research Lead Award · *Governor's STEM Scholars, Research & Development Council of New Jersey* (1000 USD)
IHFC Travel Grant · *International Heat Flow Commission* (300 USD)

2025 Dean's Honors List (Spring Quarter) · *University of California, Santa Cruz*
Excellence Fellowship · *Earth & Planetary Sciences, Rutgers University* (40,000 USD)
Eli Silver Earth & Planetary Science Opportunities Fund · *Earth & Planetary Sciences, UCSC* (2,000 USD)

2023 Marschak Astronomy SBCC Foundation Scholarship · *Earth & Planetary Sciences, SBCC* (1,000 USD)

2022 Neil Currie Award: Outstanding Planetary Petrologist · *Earth & Planetary Sciences, SBCC* (500 USD)

Total: 44,800 USD

Professional Experience

2019-2023 Lead Astronomy Programs Presenter, Coordinator
Astronomy Department & School and Teacher Services, Santa Barbara Natural History Museum (SBMNH)

Conference Presentations

Posters

2026 **Cottom, R.** & Ojha, L. (2026). Dike-mediated water transport in the Europan ice shell: Implications for the feasibility of ice-ocean exchange. *57th Lunar and Planetary Science Conference*, #1881.

2023 **Cottom, R.** & Holmbeck, E. (2023). Deriving the First Stellar Mass Function from the Light-Element Abundances of Metal-poor Stars. *Bulletin of the American Astronomical Society, 55*(2).

Teaching Assistantships

Fall 2026 01:460:201 *Earthquakes and Volcanoes*, Earth and Planetary Sciences, Rutgers University

Spring 2023 ERTH 232A *Field Study: Petrology of the Death Valley Region* (5 Days and 4 Nights), Earth and Planetary Sciences, SBCC

Other Presentations

2026 *Ice and Cryovolcanism: Mars and Beyond* · **Guest Lecturer**, 01:460:222 Planet Mars, Rutgers
Jupiter and the Galilean Satellites · **Guest Lecturer**, 01:460:224 Geology of Moons & Planets, Rutgers

2025 *Ocean Worlds and Planetary Science* · **Invited Speaker**, SBCC Astronomy Club

2023 *Algal Members of the Crystal Springs Formation* · **Tutor Talk**, Death Valley Petrology Field Course, SBCC
Galactic Archaeology · **Invited Speaker**, Santa Barbara Astronomical Unit: First Friday of the Month Speaker Event
Astronomy 101 · **Speaker**, Docent Training, Santa Barbara Museum of Natural History

2022 *Galactic Archaeology* · **Invited Speaker**, Science Pub at Dargan's, Santa Barbara Museum of Natural History
Galactic Archaeology · **Invited Speaker**, SBCC Astronomy Club

Mentoring

2026-2027 Research Lead, Governor's STEM Scholars · *High School Students*, Research & Development Council of New Jersey

2026 Research Mentor, 01:750:155 Intro. to Research and Research Communities in Physics · *Undergraduates*, Rutgers

2023-2026 Evan Grant · *Undergraduate*, University of California, Berkeley

2023-2026 CASSI Mentorship Program · *Undergraduates*, Carnegie Observatories Summer Student Internship

Organizational Roles & Outreach

2025 Committee Lead · *Commencement A/V Committee*, Earth & Planetary Sciences, UCSC

2023 - 2025 Member · *Astrobiology Reading Group*, Earth & Planetary Sciences, UCSC
Member · *Cosmochemistry Reading Group*, UCSC

2023 - 2024 Events Coordinator · *Women in Physics and Astronomy*, UCSC

2021 - 2023 President, Officer · *Astronomy Club*, Earth & Planetary Sciences, SBCC

2019 - 2023 Organizer and observatory operator · *Montly Star Party*, SBMNH, Santa Barbara, CA

2021 - 2023 Organizer and communicator · *Astronomy Club Star Parties*, Earth & Planetary Sciences, SBCC

2022 - 2023 Organizer and communicator · *STEM Day*, Earth & Planetary Sciences, SBCC

2023 Communicator · *Science Discovery Day (STEM Transfer Program)*, Earth & Planetary Sciences, SBCC

2022 Communicator · *Ask an Astronomer Table*, Carnegie Observatories, Pasadena, CA
Communicator · *AstroFest 22 (Carnegie Observatories)*, Pasadena, CA
Communicator · *La Cumbre Junior High School Science Night (SBCC Astronomy Club)*, Santa Barbara, CA

Skills

Leadership Event organizing and hosting; leading educational programs; coordinating outreach; mentoring

Coding Python, LaTeX, R, DigiStar

Software ArcGIS Pro (+ QGIS and ArcMap), COMSOL Multiphysics, iSALE, Jupyter Notebook, Microsoft Suite, Open Broadcast Software, Moog, Spectroscopy Made Hard(er), StarFit

Languages Greek (native), English (native), French (elementary)